DSA 210 Term Project
Final Report

Analyzing Personal Scuba Diving Logs from a Suunto Ocean Diving Computer

Aksel Ünlü 29540 - Spring 2026

Sabancı Üniversitesi

Motivation

As an active scuba diver and the President of the Sabancı University Subaqua Society (SUSS), I have completed over 500 dives since 2020. In 2024, I began using a Suunto Ocean diving computer, which enabled me to precisely log and export the technical parameters of my underwater activities. This project was motivated by a genuine scientific curiosity about my own diving behavior: does maximum dive depth constrain how long I stay underwater? From a physiological standpoint, deeper dives accelerate nitrogen loading in body tissues, which should theoretically force shorter bottom times to remain within safe no-decompression limits. I wanted to test whether this theoretical constraint is reflected in my actual dive logs.

Data Source

The dataset consists of 118 personal dive logs recorded by my Suunto Ocean diving computer between July 2024 and January 2026. The data was exported in bulk from the Suunto mobile application as .fit binary files; one file per dive. A custom Python script (parse.py) was written using the fitparse library to extract the following variables from each file: date, maximum depth (m), dive duration (seconds), average water temperature (°C), and minimum water temperature (°C). The parsed data was saved as a single CSV file (dives_parsed.csv) for further analysis. Raw .fit files were excluded from the repository as they contain personal data, but are available upon request.

Data Analysis

Data Cleaning

Prior to analysis, 33 dives were identified as noise and removed, leaving 85 clean dives. A dive was classified as noise if any of the following conditions were met, defined based on domain knowledge as a certified scuba diving instructor:

- Maximum depth less than 5 meters; accidental logs, pool sessions, or surface intervals
- Duration less than 8 minutes; accidental computer triggers
- Maximum depth less than 6 meters AND duration less than 20 minutes; shallow instructional dives that do not follow standard recreational profiles

Exploratory Data Analysis

Summary statistics of the cleaned dataset revealed a mean maximum depth of 21.4 meters (Range: 5.1 m - 44.5 m) and a mean dive duration of 42.5 minutes (Range: 10 min - 78.8 min). Histograms showed that depth is bimodally distributed, with peaks around 18-20 meters and 30-35 meters, reflecting the two most common recreational dive depth ranges. Duration is approximately normally distributed and centered around 40-45 minutes. A scatter plot of depth versus duration showed no visible trend.

Hypothesis Testing

Three statistical tests were applied to test the hypothesis that deeper dives have shorter durations:

- Independent samples t-test: $t = -0.058$, $p = 0.954$
- Mann-Whitney U test: $U = 847.0$, $p = 0.688$
- Spearman correlation: $r = 0.094$, $p = 0.394$

All three tests returned non-significant results at $\alpha = 0.05$. The mean duration of shallow dives (less than 20m) was 42.5 minutes, virtually identical to deep dives (42.6 minutes).

Machine Learning

Two supervised and one unsupervised ML method were applied:

Simple Linear Regression and Polynomial Regression (degree 2) were used to model dive duration as a function of maximum depth. Both models returned an R^2 score of approximately 0.00, confirming that depth has no predictive power over duration. The linear model coefficient of -0.0049 indicates that each additional meter of depth corresponds to less than one second of reduced duration; a negligible effect.

K-Means Clustering was applied to group dives by depth and duration jointly. The Elbow Method indicated an optimal k of 3. The resulting clusters were: Cluster 0 (mean depth 15.4m, mean duration 56.3 min), Cluster 1 (mean depth 16.7m, mean duration 37.1 min), and Cluster 2 (mean depth 33.6m, mean duration 42.6 min). Notably, the clusters were separated primarily by duration rather than depth, with the deepest cluster (Cluster 2) having a similar mean duration to the shallower Cluster 1.

Findings

Across all analyses, the results consistently showed no significant relationship between maximum dive depth and dive duration. The original hypothesis; that deeper dives would have shorter durations due to nitrogen loading and decompression constraints, was not supported by the data. Instead, dive duration appears to be planned and managed consistently regardless of depth. This likely reflects disciplined air management and pre-planned dive profiles typical of an experienced diver and instructor, rather than reactive adjustments to decompression limits.

Limitations and Future Work

Several limitations should be acknowledged. First, the sample size of 85 dives, while sufficient for the analyses performed, limits statistical power and generalizability. Second, air consumption data was not available in the exported .fit files, preventing direct validation of the air efficiency assumption underlying the hypothesis. Third, important confounding variables such as dive site type, current strength, visibility, and diving purpose were not controlled for. Fourth, all dives were conducted by a single experienced diver, which may limit the generalizability of findings to other diver profiles.

Future work could incorporate air consumption data by manually recording tank pressure readings from the Suunto app. Additional variables such as dive site characteristics and environmental conditions could be added as features to build a more comprehensive predictive model. Expanding the dataset to include dives from other divers of varying experience levels would also allow for a more generalizable analysis of the depth-duration relationship.

AI Assistance Disclosure

This project was developed with assistance from Claude (Anthropic). Which was used for syntax help in data parsing code and general guidance on library usage. All core decisions were made independently by me, including the project idea, hypothesis formulation, choice of data source, data cleaning rules and thresholds, interpretation of results, and selection of appropriate hypothesis tests based on course material. I also identified and resolved issues with the raw data format, determined noise filtering criteria based on personal domain knowledge as a certified diving instructor, and drew all analytical conclusions from the results.